

Sanitation: Terra Preta Toilets

Carpathian OER Commons — Utilities Module

Michel Garand | 2026-09-29 | Version v0.1 | CC BY-SA 4.0

Maturity: L1 — designed and costed, untested. Every cost line is an estimate or a listed price, labelled as such in Section 6. No cubicle has been built and no bucket filled. Read every figure below at this level and no higher.

Governed by Section 2 of the compendium. No child appears in this module in any form. The site is not named; site plans are schematic; photographs show structures and systems only. Design loads are site capacity, used for sizing only.

Converted from the sanitation working paper, EN v0.2, Sections 1 to 3, 8.1, 9, and 10. The toilets end where the full bucket leaves the cubicle: the exchange, the rinse point, and everything after belong to the terra preta module; the two existing flush toilets and their septic tank to the greywater, reed bed, and pond module; the building to the living layer.

Permits and local rules. This module is a guide, used in different regions and countries under different laws and guidelines. Depending on where it is built, at what scale, and for whom, permits, registrations, or approvals may be required — for building, water, sanitation, fire, waste, food, or the use of forest and land. Section 10 records what was found for Ukraine, dated, and unchecked until signed; a reader elsewhere checks their own rules with the competent authorities before building, and nothing here replaces that.

1. Purpose

Toilets for everyone on site at capacity that need no water, no pipe, and no power to work: a seat over a lidded bucket in a warm, private, lit cubicle, a scoop of char and carbon cover after every use, a vent, and handwashing at the door. Simple where it is handled; everything difficult happens afterwards, in the terra preta module.

Five rules. A bucket under a seat: no separator, no moving part, no plumbing; urine goes in with the rest, as decided on 2026-09-29. Char and cover at every use: a scoop from the bin beside the seat, so that no deposit is left uncovered. Air out, not in: a dark vent pipe from the bucket chamber above the roof, with a fly screen and a small fan. Hands washed at the door: soap, water, and a towel within reach of every cubicle. Clean twice a day at least, by a named adult, paid, with a written procedure; toloka is not the name for this work.

Connections. Depends on: the biochar module for char, and the terra preta module for the cover it sizes; the energy module for the fans on the essential circuit, or a small panel of their own; the washroom block module for washbasins and hot water; the written answer of the district sanitary service on whether bucket toilets are accepted for the children's side. Feeds: the terra preta module, with the full buckets every day. Beside it: the two existing flush toilets, which carry about a third of uses and drain to the septic tank of the greywater module.

Patterns. Grows from: not yet named. How homesteads in this valley kept a privy, and where it stood, are questions for the elders; the literature found nothing. Feeds: not yet named. Resonates with: 178 Compost — toilets that feed the soil, through the terra preta module, for the trees; 127 Intimacy Gradient — the cubicles off the common rooms, private and reached without passing through them; 36 Degrees of Publicness — a unit for staff and visitors apart from the children's side. In tension with: 129 Common Areas at the Heart — the toilets near the shared rooms, never opening onto them; 120 Paths and Goals — the path from the sleeping rooms to the toilets at night, lit and short, with the fire protection and night lighting module.

2. Context of validity

Condition	Range this design holds for
Climate	[set] — cubicles inside a heated building
Altitude	[set]
Group size	capacity: 34 people, the example below
Uses	toilet_uses: 5 a person a day
Share	bucket_share: two-thirds of uses; flush toilets the rest
Cubicles	six, as below
Power	none to work; the fans on the essential circuit
Rules	bucket toilets accepted by the sanitary service

If the sanitary service requires plumbed units for the children's side (Section 10), the bucket toilets serve staff, guests, and the outdoor working day, bucket_share falls to their share, [set], and the cubicles in Section 4 fall with it. The fallback is an extension of the existing flush toilets, not a redesign (sanitation paper, Section 9).

3. Sizing method

What you must measure yourself before using this module:

- **Your uses** — uses a person a day and the share the flush toilets take, from the first sessions.
- **Your cover** — how much a use takes to leave no smell, from the first week.

- **Your building** — where the cubicles can stand inside the heated envelope, with a vent route to the roof.
- **Your rules** — the written answer of the sanitary service.

Our example: capacity, 34 people; toilet_uses, 5 a day; bucket_share, two-thirds; urine_per_person, 1.4 L; faeces_per_person, 0.128 kg; char_at_seat, 0.5 L; cover_at_seat, 0.8 L, [assumed]. Values from the site parameters file, held in the workshop.

Quantity	Calculation	Our example
Uses at the buckets, a day	$34 \times 5 \times \text{two-thirds}$	about 113
Uses a cubicle, a day	$113 \div 6$	about 19
Bucket volume, a day	the terra preta module, 3.1	about 64 L
A bucket, a day	$64 \text{ L} \div 6$	about 11 L
Char a use	$11.3 \text{ L} \div 113$	about 0.1 L
Cover a use	$18.1 \text{ L} \div 113$	about 0.16 L
Char and cover, a seat a day	$29.4 \text{ L} \div 6$	about 4.9 L
A 60 L bin lasts	$60 \text{ L} \div 4.9 \text{ L}$	about 12 days
Buckets	6 in use, 6 clean	12 of 30 L

A 30 L bucket exchanged daily carries about 11 L, and never gets heavy enough to be a lifting problem. The paper's two 30 L buckets changed daily at 25 people counted faeces and char only; with urine and cover the load rises, and daily exchange of every bucket keeps it light.

Six cubicles, and why. The paper has four on the children's side, split between boys and girls, one for staff and visitors, and one that is accessible. The repealed sanitary rules for children's recreation establishments gave one toilet bowl for every 15 to 20 users (ДСанПиИ 5.5.5.23-99, point 5.5); six cubicles and the two flush toilets are well inside that at capacity. What the 2026 regulation asks is for the enquiry to confirm (Section 10).

The vent. Jenkins holds that cover of the right kind, in the right amount, makes a toilet odourless "without the need for venting" (Humanure Handbook, cover material guidance). The repealed rules gave toilets their own exhaust ventilation (point 5.7), and the 2026 rules are not yet read in full; so each cubicle keeps a vent — a pipe of at least 110 mm, rising more than 300 mm above the roof, painted dark, with an aluminium fly screen of 1.2 to 1.5 mm mesh (Tilley and colleagues, Compendium of Sanitation Systems and Technologies, 2014) — and a 12 V fan.

4. Bill of materials

Quantities for the example. Alternatives are named in the design; none is yet known to work here.

Item	Quantity	Alternatives in the design
Seat box, plywood, with seat and lid	6	a ready peat toilet
Bucket, 30 L, food-grade, lidded	12	33 L
Char-and-cover bin, 60 L, lidded	6	none named
Scoop	6	none named
Vent pipe, 110 mm, fly screen	6 runs	125 mm
Fan, 12 V, 100 mm	6	one fan on a manifold
Cubicles, opaque walls and doors	6	HPL partitions
Accessible cubicle fittings	1 set	none named
Grab rail and fold-down rail	1 each	none named
Paper holder	6	none named
Soap and towel dispensers	2 each, [assumed]	hand dryer
Washable wall and floor finish	[set]	PVC panels; tile

The cubicles, as designed. Warm, private, and lit, inside the building; opaque walls and doors; washable surfaces; a seat box with a hinged front for the bucket, the lid closing over it; the bin of char and cover beside the seat with its scoop; the vent from the bucket chamber; handwashing at the door. The children's side split between boys and girls; a unit for staff and visitors apart.

The accessible cubicle. To ДБН В.2.2-40:2018 with its Amendment 2, as summarised by the ministry: at least 1.65 × 1.8 m in a reconstruction, 1.7 × 2.2 m new; a door of 0.9 m without threshold, opening outward; two grab rails 70 to 75 cm apart, one of them fold-down; a seat 40 to 48 cm high; an alarm within reach from the seat and from the floor.

An outdoor unit, if one is used for the working day. It keeps the distances the regulation gives outdoor toilets — at least 50 m from sleeping buildings and the dining hall, as reported — and those of the compost and terra preta modules from the spring, the stream, and the pond. None has been measured.

5. Build

Step	Trade	Time	If done out of order
Written enquiry answered	household	[from L2]	cubicles refused
Layout, vent routes	household	[from L2]	vents through rooms
Walls, doors, finishes	carpenter	[from L2]	[from L2]
Vent pipes, roof sleeves	roofer	[from L2]	leaks, smell
Fans on the circuit	electrician	[from L2]	no draught
Seat boxes, seats, bins	carpenter	[from L2]	[from L2]
Accessible fittings, alarm	fitter	[from L2]	[from L2]
Written procedure	household	[from L2]	no answer at inspection

6. Cost

Estimates, not quotations: each line is either a price listed by a Ukrainian retailer, seen on 2026-09-29, or the sanitation paper's own estimate for autumn 2026, marked "paper"; the paper's lines were for six cubicles, as here. Place: not a quotation; national listings. Exchange rate: exchange_rate, 52 UAH to the euro, the National Bank's mid-September 2026 rate, rounded.

6.1 Toilets

Item	UAH	EUR	Basis
Six seat boxes, seats and lids	24,000	462	paper
Buckets, 12 × 30 L at 336	4,032	78	estimated, listing
Bins, 6 × 60 L at 637	3,822	74	estimated, listing
Scoops, 6 at 29	174	3	estimated, listing
Subtotal	32,028	616	

6.2 Ventilation

Item	UAH	EUR	Basis
Vent pipes and extract, six	15,000	288	paper
Subtotal	15,000	288	

The paper's line against listings: a 2 m length of 110 mm pipe at 311 UAH and a 12 V fan at 816 UAH, seen 2026-09-29; the paper's figure is kept because it includes the roof sleeves and the work.

6.3 Cubicles

Item	UAH	EUR	Basis
Cubicle fit-out, doors, finishes	25,000	481	paper
Accessible cubicle fittings	15,000	288	paper
Paper holders, 6 at 91	546	10	estimated, listing
Soap dispensers, 2 at 93	186	4	listing; qty assumed
Towel dispensers, 2 at 590	1,180	23	listing; qty assumed
Subtotal	41,912	806	

The two rails for the accessible cubicle are listed at 2,345 and 2,435 UAH, inside the paper's line.

6.4 Summary

Section	UAH	EUR
Toilets	32,028	616
Ventilation	15,000	288
Cubicles	41,912	806
Fit-out subtotal	88,940	1,710
Design and sanitary review, 8 percent	7,100	137
Contingency, 20 percent	17,800	342
Total external cost	113,840	2,189

Against the paper's 86,200 UAH for the same six cubicles, the difference is the bins for char and cover and the dispensers. The building is not costed here. The percentages are the paper's own.

Contribution, not in the total. None counted.

Running cost. Cleaning at least twice a day: disinfectant at a listed 412 UAH for 5 L of chlorine solution, quantity [set]; paper, soap, and towels, [set]. Char and cover come from the biochar and terra preta modules. The cleaning is paid work, [set] hours, at no less than the legal minimum wage, 8,647 UAH a month in 2026, for the hours worked.

7. Operation and maintenance

Responsible: one named adult, paid, with a deputy — [from L2]. Not a rota.

- **At every use:** a scoop of char and cover over the deposit; the lid closed.
- **Twice a day at least:** cubicles cleaned with detergent and disinfectant, seats, handles, and floors included, and the cleaning recorded; bins topped up; paper, soap, and towels checked.
- **Daily, in session:** every bucket exchanged by the responsible adult, as the terra preta module describes, and a clean bucket set in with a layer of cover in its base.
- **Weekly:** vent screens and fans checked; bins refilled from the store.
- **Seasonal:** vent pipes cleared; roof sleeves checked.

The responsible adult wears gloves and is trained in the written procedure, as WHO asks of anyone who handles excreta (Guidelines on Sanitation and Health, 2018). How the programme describes the toilets to those who use them is for the programme modules, in standing format.

8. Failure modes

From the design, not from experience.

Failure	Early sign	Repair and cost
Smell in the cubicle	smell	more cover; fan checked
Flies	flies at the seat	cover; screen; lid closed
Bucket too full	weight	exchange more often
Bin empty	[from L2]	refill daily in session
Fan stops	no draught	power; spare fan, 816 UAH

Failure	Early sign	Repair and cost
Vent leaks at roof	stain	sleeve resealed, [from L2]
Toilets refused	written answer	the plumbed fallback

The paper names the four questions an inspector will ask — smell, flies, hand hygiene, and what happens to the material — and answers them with the cover, the vent, handwashing at the door, and the terra preta module's written procedure and rule of the year.

9. What we got wrong

Nothing yet: this module has not been built.

10. Regulatory notes

Country	Instrument, number, date	Checked by, date
Ukraine	found 2026-09-29, not checked — see below	—
Romania	not checked	—
Poland	not checked	—
Slovakia	not checked	—
Hungary	not checked	—
Serbia	not checked	—
Czechia	not checked	—

Ukraine, found and not checked. Instruments found on 2026-09-29 and read on secondary copies and official summaries, because the statute site could not be reached; each is read on zakon.rada.gov.ua and signed before this row counts as checked.

- **MOH Order No. 377 of 20.03.2026**, the sanitary regulation for children's health and recreation establishments, registered 20.04.2026 as No. 547/45941, for every such establishment whatever its ownership, tent camps excepted; read in official summaries only. Sanitary units cleaned at least twice

a day with detergents and disinfectants; outdoor toilets at least 50 m from sleeping buildings and the dining hall; premises adapted for persons with disabilities. Its fixture ratios, the separation of units, and whether a toilet without sewerage is allowed indoors are not verified.

- **ДСанПіН 5.5.5.23-99**, repealed by Order No. 377, for comparison: additional toilets on the grounds sewered (point 3.9); one bowl for every 15 to 20 users and one washbasin for every 8 (point 5.5); toilets with their own exhaust ventilation (point 5.7).
- **МОН Order No. 89 of 07.02.2012**, for tent camps: one toilet for every 20, separate units, washbasins beside them, and composting toilets named as the best option there (point 7.7).
- **МОН Order No. 145 of 17.03.2011**, point 2.23: mobile toilets and washbasins at mass events; nothing on dry or composting toilets.
- **ДБН В.2.2-40:2018**, accessibility, with Amendments 1 and 2: the accessible cubicle as in Section 4; the point numbers differ between sources and are to be read in the code.

No rule found explicitly allows or forbids an indoor bucket toilet. The written enquiry to the district office of the state food-safety and consumer-protection service, with the design drawn and the procedure written, asks whether bucket toilets are accepted for the children's side, which establishment category the programme falls under, and what Order No. 377 asks of the units; the plumbed fallback in Section 2 is the answer if they are not.

11. Adaptation notes

The design is a seed, not a copy: a different place grows a different form from it.

- **A drier valley:** unchanged; the toilets use no water.
- **A colder valley:** the cubicles stay inside the heated envelope; vent pipes insulated where they cross cold roof space.
- **A smaller group:** fewer cubicles, keeping a unit for staff and visitors and an accessible one.
- **A larger group:** more cubicles at the ratio your rules give; the terra preta module's vessels scale with them.
- **Plumbed units required:** the bucket cubicles serve staff, guests, and the working day; the children's side extends the flush toilets.
- **No power:** the vent runs on draught alone, dark and tall; the fan is an aid, not a condition.

12. Provenance and licence

- **Written by:** Michel Garand. [set — others as they choose to be named]

- **From whose knowledge:** the sanitation working paper, EN v0.2; Joseph Jenkins, The Humanure Handbook, for the bucket toilet under cover; Tilley and colleagues, Compendium of Sanitation Systems and Technologies, Eawag, 2014, for the vent; WHO, Guidelines on Sanitation and Health, 2018, for handwashing and for those who handle excreta; the Ukrainian instruments in Section 10; Ukrainian retail listings of 2026-09-29 for prices.
 - **Traditional knowledge:** none. How homesteads here kept a privy is for the elders; the literature found nothing.
 - **Text:** CC BY-SA 4.0
 - **Drawings and designs:** [hardware licence — open decision]
 - **Measurement data:** none yet
 - **Gate 1:** household decision, 2026-09-29, recorded by Michel Garand — v0.1 may be published as it stands, at L1, in English, Ukrainian, and German
 - **Wartime review:** 2026-09-29, Michel Garand — v0.1 read against compendium Section 2; nothing places the site; no child is counted; the distances are the instruments' minimums, not measurements of the site; accepted
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13. What to check

- **The written answer** — whether bucket toilets are accepted for the children's side, the establishment category, and what Order No. 377 asks of the units; its full text read.
 - **bucket_share** and **toilet_uses** — from the first sessions.
 - **cover_at_seat** — from the first week.
 - **The vent routes** — from the building survey.
 - **The accessible cubicle** — the code read, its point numbers settled.
 - **The dispensers** — how many hand-wash points the layout gives.
 - **The finishes** — area and material, priced.
 - **Cleaning** — disinfectant, paper, soap, towels, and hours, set.
 - **The Ukrainian row** — every instrument read and signed.
 - **Quotations** for the lines in Section 6, dated, to replace the listings and the paper's estimates.
 - **From L2, once built:** build times and out-of-order consequences in Section 5; the cleaning record in Section 7; early signs and repair costs in Section 8; Section 9.
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